Model-reference Adaptive Systems (Åström & Wittenmark, Examples 5.1, 5.2 and 5.7)

Adaptive Control, Homework 5

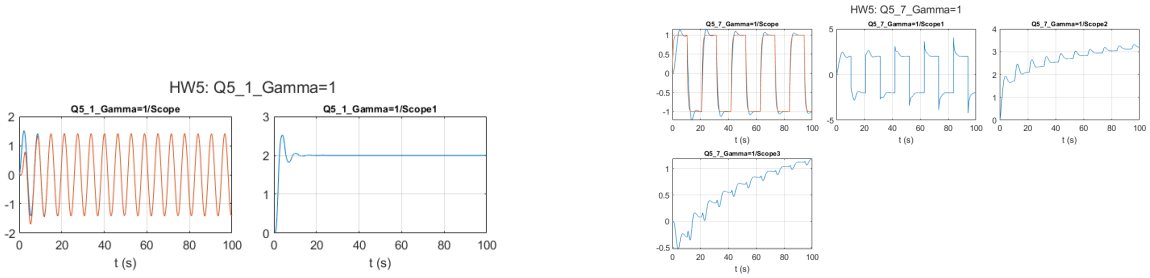
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All cases are subsystems of `HW5.slx`. Each was simulated for three adaptation gains γ and compared with the book.

Example 5.1: MIT rule, adapting a feed-forward gain. $\gamma = 0.5, 1, 2$. In the re-run, the gain estimate settles in every case; with larger γ it moves faster and oscillates more before settling.

Example 5.2: MIT rule for a first-order plant with two parameters. $\gamma = 0.2, 1, 5$. I plotted the output, the control signal and the estimates θ_1, θ_2.

Example 5.7: Lyapunov-based MRAS for the same plant. $\gamma = 0.2, 1, 5$. The update law is derived from a Lyapunov function instead of the MIT gradient rule.



Re-run in R2023b: example 5.1 (left) and example 5.7 (right), both with $\gamma = 1$.